A SIGNED-DIRECTION INVARIANT FOR TRIANGLE TILINGS, AND THE EXCLUSION OF PRIMES CONGRUENT TO 3 MODULO 4

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Abstract. For a triangle T and a triangle ABC, an *N-tiling* is a dissection of ABC into N triangles each congruent to T. Erdős asked which N occur; a folklore conjecture is that no prime $N \equiv 3 \pmod 4$ does. By the classification of Laczkovich and the branch theorems of Beeson, the conjecture reduces to a tile with a $2\pi/3$ angle on a non-equilateral triangle, where a prime count forces ABC isosceles. For that isosceles case Beeson established a finite lower bound on N, and Zhang's recent constructions led to the conjecture that no prime occurs; we prove it. We introduce a translation-invariant, signed-direction tiling functional, linear in edge length and hence insensitive to non-edge-to-edge incidences, whose integrality forces, for an isosceles target, the elementary fact that $(c - a - b)/\sqrt{b}$ is never an integer for a primitive triple with $c^2 = a^2 + ab + b^2$. With a self-contained reduction of the scalene shapes, this completes the exclusion of primes $\equiv 3 \pmod 4$, conditional on the cited classification. The number-theoretic core is machine-checked in Lean 4; the geometric lemmas are verified numerically and are offered for refereeing.

1. Introduction

Let T be a triangle (the *tile*). An *N-tiling* of a triangle ABC is a way of writing ABC as a union of N triangles, each congruent to T, with pairwise disjoint interiors. We do *not* assume the tiling is edge-to-edge: a vertex of one tile may lie in the relative interior of an edge of another. Erdős asked (reported by Soifer [12]) to determine the set of N for which some triangle admits an N-tiling by some tile. Every perfect square occurs (any triangle, by the n^2 subdivision), and Soifer observed that $n^2 + m^2$, $2n^2$, $3n^2$ and $6n^2$ occur as well. The outstanding question is whether the primes $\equiv 3 \pmod 4$, which are exactly the primes that are not sums of two squares, are excluded.

Theorem 1. *No prime $N \equiv 3 \pmod 4$ is a number of congruent triangles into which some triangle can be cut. In particular, no triangle can be cut into 19 congruent triangles.*

The proof rests on Laczkovich's classification of triangle tilings [9, 10], organized by Beeson across a series of papers [1, 2, 3, 4, 5]. For a prime number of tiles every branch of the classification is excluded by an existing theorem, with one exception: a tile having a $2\pi/3$ angle, tiling a non-equilateral triangle. We treat that branch.

This branch is the active frontier. Zhang [7] classifies the non-equilateral $2\pi/3$ shapes into six sporadic similarity types (Section 2 below), constructs families of tilings for each, and conjectures that the constructed N, all composite, are the only ones; the present results prove the no-prime half of that conjecture for the isosceles type, and reduce the scalene types directly. For the isosceles type Beeson [3] had already established a finite lower bound $N \geq 2736$, excluding small N; the invariant below removes the bound, excluding *every* prime. The adjacent Erdős Problem 633, on tilings into a square number of triangles, was recently settled by Beeson, Laczkovich and Zhang [8].

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Two reductions prepare it. First, by the rationality theorem of Beeson and Zhang [6], such a tile has commensurable sides, so after scaling we may take it integer-sided with $c^2 = a^2 + ab + b^2$. Second, an elementary area-and-incidence argument shows that a *prime* number of such tiles forces ABC to be *isosceles* (Section 3).

The heart of the paper is Section 4. The natural parity argument for a $2\pi/3$ tile (a black/white colouring of the tiles) fails, because a vertex of the tiling may be surrounded by an even number of tiles, so no consistent two-colouring exists. We replace it by a functional Φ that assigns to a directed edge of direction $\theta = j\frac{\pi}{3} + k\alpha$ the weight (length) $\cdot\, (-1)^j$. Two features make it work: $\Phi(\theta + \pi) = -\Phi(\theta)$, so interior edges cancel; and the weight is *linear in edge length*, so the cancellation is unaffected by non-edge-to-edge incidences, where one long edge meets several shorter ones summing to it. Summing over the tiles, Φ of the boundary equals an integer multiple of a fixed tile-value $c + a - b$. For an isosceles target this forces $(c - a - b)/\sqrt{b} \in \mathbb{Z}$, which never holds (Section 5). Assembling the branches proves Theorem 1 (Section 6).

The combinatorial and numerical claims are verified by the scripts in the accompanying repository, and the number-theoretic core (Lemma 6) is formalized and machine-checked in Lean 4 with Mathlib; see Section 7.

2. Preliminaries

The classification. By Laczkovich [9, 10], every N-tiling of a triangle falls into one of finitely many types of pairs (ABC, T). For a prime N the relevant dichotomy, in Beeson's organization [4], is: either the angles of T are pairwise commensurable with π, or they are not. If they are commensurable, then N is a square, a sum of two squares, or 2, 3 or 6 times a square [4, Thm. 7]; a prime $\equiv 3 \pmod 4$ is none of these. If they are not, then (after the right-angle and similar-tile cases, which give N a sum of two squares or $N = n^2$ [1, 11]) the tile has a special angle and falls into one of: $3\alpha + 2\beta = \pi$, for which N is not prime [2]; an isosceles tile with $\gamma = 2\alpha$, for which N is not squarefree [3, Thm. 11.7]; or the tile has an angle equal to $\pi/3$ or $2\pi/3$. The case of a $\pi/3$ or $2\pi/3$ tile on an *equilateral* triangle gives N not prime [5, Thms. 3 and 6]. What remains for a prime N is a tile with a $\pi/3$ or $2\pi/3$ angle on a *non-equilateral* triangle. A vertex enumeration (Section 2) shows the $\pi/3$ case admits only equilateral or tile-similar ABC, already covered; so the sole remaining branch is a $2\pi/3$ tile, which we now fix.

The tile. Throughout, T has angles (α, β, γ) opposite sides (a, b, c), with $\gamma = 2\pi/3$, hence $\alpha + \beta = \pi/3$; we treat the incommensurable case, α an irrational multiple of π. By Beeson–Zhang [6], T has commensurable sides, so we scale to

$$(a, b, c) \in \mathbb{Z}^3, \qquad \gcd(a, b, c) = 1, \qquad c^2 = a^2 + ab + b^2 \tag{1}$$

(the law of cosines at $\gamma = 2\pi/3$). We call such a triple a *primitive* 120°*-triple*. If a prime p divides two of a, b, c it divides the third (by (1)), so (1) gives $\gcd(a, b) = \gcd(a, c) = \gcd(b, c) = 1$. One computes

$$\sin\alpha = \tfrac{a\sqrt{3}}{2c}, \quad \cos\alpha = \tfrac{2b+a}{2c}, \qquad \sin\beta = \tfrac{b\sqrt{3}}{2c}, \quad \cos\beta = \tfrac{2a+b}{2c}. \tag{2}$$

The direction grid. Set $G = \{\, j\frac{\pi}{3} + k\alpha : j, k \in \mathbb{Z} \,\} \pmod{2\pi}$. We claim every edge of every tile in a tiling points in a direction lying in G, for a single fixed reference. Fix the reference along an edge of one tile; its three edge-directions are then in G, since the tile angles $\alpha, \beta = \frac{\pi}{3} - \alpha, \gamma = \frac{2\pi}{3}$ all lie in $\mathbb{Z}\alpha + \mathbb{Z}\frac{\pi}{3}$. Now argue by adjacency. The tiling fills a connected region, so any tile t' shares at least a boundary point P with the union of the tiles already placed; at P the surrounding corner angles each belong to $\{\alpha, \beta, \gamma\}$, and a directed edge of t' at P differs from a known grid direction at P by a sum of such angles, hence lies in G; the remaining edges of t' then differ from it by $\mathbb{Z}$-combinations of $\alpha, \frac{\pi}{3}$, so all of t' is in G. This uses only the angles meeting at a point and so applies verbatim at non-edge-to-edge incidences, where P lies in the relative interior of an edge (the angles there sum to π rather than 2π). Since α/π is irrational, $\frac{\pi}{3}$ and α are $\mathbb{Q}$-linearly independent, so the pair (j,k) is determined by the direction (and $(-1)^j$ by the direction alone, as $2\pi = 6\cdot\frac{\pi}{3}$).

Vertex types and shapes of ABC. A corner of ABC is filled by tile-corners, so its measure is $m\alpha + k\frac{\pi}{3}$ with $(m,k) = (P-Q,\, Q+2R)$ for nonnegative integers P,Q,R counting the α,β,γ corners there. The three corners satisfy $\sum m_i = 0$ and $\sum k_i = 3$. Enumerating all such triples (a finite computation; see Section 7) gives exactly eleven shapes of ABC: the equilateral triangle; two isosceles triangles, with base angles α (apex $\alpha+3\beta$) and β (apex $\beta+3\alpha$); and eight scalene triangles, namely

$$F_1 = (\alpha, \alpha+\beta, \alpha+2\beta),\quad F_2 = (2\alpha, 2\beta, \alpha+\beta),\quad F_3 = (\alpha, 2\alpha, 3\beta),\quad F_4 = (\alpha, 2\beta, 2\alpha+\beta),$$

their reflections (interchanging $\alpha \leftrightarrow \beta$), and the tile-similar triangle $(\alpha, \beta, 2\pi/3)$.

The area identity. If ABC is similar to a fixed shape whose primitive integer side-proportion vector is D (gcd of its entries 1), and ABC is N-tiled, then each side of ABC is a disjoint union of whole tile-edges (a tile meeting a straight portion of the boundary has a whole side on it), hence an integer; so the sides of ABC are $k\cdot D$ for an integer $k\ge 1$, and

$$N = \frac{\mathrm{Area}(ABC)}{\mathrm{Area}(T)} = k^2 N_0, \qquad N_0 := \frac{\mathrm{Area}(D)}{\mathrm{Area}(T)}, \qquad \mathrm{Area}(T) = \tfrac{\sqrt{3}}{4}ab. \tag{3}$$

This is a necessary condition on N.

3. Reduction to the isosceles case

Proposition 2. *If a $2\pi/3$ tile N-tiles a triangle and N is prime, then ABC is isosceles.*

Proof. By Section 2, a non-isosceles ABC is equilateral, tile-similar, or one of $F_1,\dots,F_4$ or their reflections. The equilateral case gives N not prime [5, Thm. 6]; the tile-similar case gives $N = n^2$ [1, 11], not prime. For the remaining shapes we compute N_0 from (3); in each case $N = k^2N_0$ is composite.

F_2 and F_4. The side-proportion vector of F_2 is $\big(a(a+2b),\, b(2a+b),\, c^2\big)$ with the $\pi/3$ vertex between the first two sides, giving $N_0 = (a+2b)(2a+b)$. For F_4 one checks the side-vector $\big(ac,\, b(2a+b),\, (a+b)c\big)$ has gcd 1 (any common prime g divides c and then $g^2 \mid 3b^2$, forcing $g=1$), giving $N_0 = (a+b)(2a+b)$. Both are products of two integer factors each at least 3, hence composite, so $N = k^2N_0$ is composite.

F_3. Clearing the common factor c, the primitive side-vector of F_3 is $\big(ac^2,\, a(a+2b)c,\, 3ab(a+b)\big)$, and $N_0 = 3(a+b)(a+2b)$. This is divisible by 3, hence composite, so $N \equiv 0 \pmod 3$ is composite.

F_1. The side-vector is $(a, c, a+b)$ with the $\pi/3$ vertex between the sides a and $a+b$, so $N_0 = (a+b)/b$ and $N = k^2(a+b)/b$. As $\gcd(b, a+b) = 1$, integrality forces $b \mid k^2$, so $N = t(a+b)$ with $t = k^2/b \in \mathbb{Z}_{\ge 1}$. It remains to show $a+b$ is composite. Suppose $a+b = p$ is prime. Since $(a+b)^2 - c^2 = ab$ by (1), we have $ab = p^2 - c^2$, so a, b are the roots of $x^2 - px + (p^2 - c^2)$, whose discriminant $4c^2 - 3p^2$ must be a perfect square d^2; then $3p^2 = (2c-d)(2c+d)$. With $2c - d < 2c + d$ the factorizations of $3p^2$ into two positive factors are $(1, 3p^2)$, $(3, p^2)$, $(p, 3p)$, and in each the smaller root $\frac{p-d}{2}$ is not a positive integer: in the first, $d = (3p^2-1)/2 > p$, so the root is negative; in the second, $d = (p^2-3)/2$ and the root $\frac{p-d}{2} = -\frac{(p-3)(p+1)}{4}$ is negative for $p > 3$; in the third, $d = p$ gives $c = p$ and $ab = p^2 - c^2 = 0$. (The case $p \le 3$ is impossible since $a+b \ge 8$ for a primitive 120°-triple.) Hence $a+b \ge 8$ is composite, and $N = t(a+b)$ is composite.

The reflected shapes are handled by $\alpha \leftrightarrow \beta$ (which swaps $a \leftrightarrow b$ and leaves each N_0 above of the same form). In every non-isosceles non-equilateral case N is composite, so a prime N forces ABC isosceles. $\square$

4. The signed-direction invariant

Fix the reference so the direction grid is $G = \{\, j\frac{\pi}{3} + k\alpha \,\}$. Because $\beta = \frac{\pi}{3} - \alpha$, every grid direction also has the form $(j+k)\frac{\pi}{3} - k\beta$, so we may read off the coefficient of $\frac{\pi}{3}$ in two coordinate systems. Define, for a direction $\theta \in G$,

$$f_\alpha(\theta) = (-1)^j \quad (\theta = j\tfrac{\pi}{3} + k\alpha), \qquad f_\beta(\theta) = (-1)^{j'} \quad (\theta = j'\tfrac{\pi}{3} + k'\beta).$$

Both are well defined (Section 2), and both satisfy

$$f(\theta + \pi) = -f(\theta), \tag{4}$$

since adding $\pi = 3 \cdot \frac{\pi}{3}$ increases the $\frac{\pi}{3}$-coefficient by 3. They are genuinely different functions: $f_\beta = (-1)^k f_\alpha$.

For $f \in \{f_\alpha, f_\beta\}$ and a directed edge of length L and direction θ, assign the weight $L\, f(\theta)$. For a tile t with boundary traversed counterclockwise, put $C_f(t) = \sum_{\text{edges}} L\, f(\theta)$, and for the boundary of ABC put $\Phi_f(\partial ABC) = \sum_{\text{sides}} L\, f(\theta)$ (counterclockwise).

Lemma 3 (Cancellation)**.** *For each $f \in \{f_\alpha, f_\beta\}$, $\displaystyle\sum_{tiles} C_f(t) = \Phi_f(\partial ABC)$.*

Proof. Group the contributions by location. Every point of ABC interior to the tiling lies on a maximal straight segment shared by the tiles on its two sides; along that segment the tiles on one side present directed edges of total length L in direction θ, and the tiles on the other side directed edges of total length L in direction $\theta + \pi$ (each side covers the segment exactly once, though the two subdivisions need not agree). By additivity in length, their combined weight is $L\, f(\theta) + L\, f(\theta + \pi) = 0$ by (4), the value f being constant along a fixed direction. This holds verbatim where a single long edge meets several shorter edges that sum to it (a non-edge-to-edge incidence): the several short weights add up to $L\, f(\theta + \pi)$. Only the boundary of ABC survives. $\square$

Lemma 4 (Tile value)**.** *For every placement of T, $C_{f_\alpha}(t) = \pm(c+a-b)$ and $C_{f_\beta}(t) = \pm(c+b-a)$.*

Proof. Counterclockwise, the three edges have lengths c, a, b (opposite γ, α, β) and their directions differ by the exterior angles $\pi - \beta = \frac{2\pi}{3} + \alpha$ and $\pi - \gamma = \frac{\pi}{3}$; so their α-frame $\frac{\pi}{3}$-coefficients are j_0, $j_0 + 2$, $j_0 + 3$. Hence under f_α the signs are $(-1)^{j_0}(+,+,-)$ (the b-edge, after the γ-turn, is the odd one out), and $C_{f_\alpha} = (-1)^{j_0}(c + a - b)$. A reflection reverses the traversal and negates C, and every placement is a grid rotation $m\alpha + n\frac{\pi}{3}$ (which multiplies C_{f_α} by $(-1)^n$) possibly composed with a reflection; so $C_{f_\alpha} = \pm(c + a - b)$ always. The same computation in the β-frame makes the a-edge odd, giving $C_{f_\beta} = \pm(c + b - a)$. $\square$

Corollary 5 (Integrality)**.** *Each of $\dfrac{\Phi_{f_\alpha}(\partial ABC)}{c + a - b}$ and $\dfrac{\Phi_{f_\beta}(\partial ABC)}{c + b - a}$ is an integer. They are separate necessary conditions: each follows from Lemma 3 for its own f, so a single non-integer value forbids a tiling.*

Proof. By Lemmas 3 and 4, $\Phi_{f_\alpha}(\partial ABC) = \sum_t C_{f_\alpha}(t) = (c + a - b) \sum_t \varepsilon_t$ with $\varepsilon_t \in \{\pm 1\}$, and likewise for f_β. $\square$

5. The isosceles obstruction

Lemma 6 (Non-integrality)**.** *Let a, k be positive integers with $\gcd(a,k) = 1$, put $b = k^2$, and let $c > 0$ satisfy $c^2 = a^2 + ab + b^2$. Then $k \nmid (a + b - c)$.*

Proof. From $c^2 = a^2 + ab + b^2$ we get $c > a$ and, since $c^2 < (a + b)^2$, also $c < a + b$. Suppose $k \mid (a + b - c)$. As $b = k^2 \equiv 0 \pmod{k}$, this gives $c \equiv a \pmod{k}$; with $0 < c - a < b = k^2$ we may write $c = a + kt$, $1 \leq t \leq k - 1$. Substituting into $c^2 = a^2 + ak^2 + k^4$ and dividing by k,

$$2at + kt^2 = ak + k^3, \qquad \text{that is} \qquad a(2t - k) = k(k - t)(k + t).$$

Then $k \mid a(2t - k)$, and $\gcd(a, k) = 1$ gives $k \mid (2t - k)$, hence $k \mid 2t$. But $2 \leq 2t \leq 2k - 2$ contains no multiple of k other than k itself, so $2t = k$. The left side is then $a(2t - k) = 0$, while the right side is $k \cdot \frac{k}{2} \cdot \frac{3k}{2} = \frac{3k^3}{4} > 0$, a contradiction. Hence $k \nmid (a + b - c)$. ☐

Theorem 7. *No prime number of $2\pi/3$ tiles tiles an isosceles triangle.*

Proof. An isosceles ABC has its base angle equal to a tile acute angle (its corner angles lie in $\mathbb{Z}\alpha + \mathbb{Z}\frac{\pi}{3}$, and the base angle is acute). We use the invariant generated by the base angle, so that the equal sides land on $\frac{\pi}{3}$-coefficient 3 (odd) and pick up $f = -1$.

If the base angle is α, then $ABC = k(c, c, 2b + a)$ (equal sides kc, base $k(2b + a)$): the base is at direction 0 ($f_\alpha = +1$) and the equal sides at $\pi - \alpha = 3\frac{\pi}{3} - \alpha$ ($f_\alpha = -1$), so $\Phi_{f_\alpha}(\partial ABC) = k(2b + a) - 2kc = k(2b + a - 2c)$. The tiling count is $N = k^2(a + 2b)/b$; since N is prime and $\gcd(a + 2b, b) = 1$, we have $k^2 = b$, i.e. $k = \sqrt{b}$. By Corollary 5 and $c^2 = a^2 + ab + b^2$,

$$\frac{k(2b + a - 2c)}{c + a - b} = \frac{c - a - b}{k} \in \mathbb{Z}, \qquad k = \sqrt{b},$$

so $k \mid (a + b - c)$, contradicting Lemma 6. If the base angle is β, then $ABC = k(c, c, 2a + b)$, the equal sides are at $\pi - \beta$, and using f_β (for which the equal sides have $\frac{\pi}{3}$-coefficient 3) the same computation gives, with $k = \sqrt{a}$, $(c - a - b)/k \in \mathbb{Z}$, again contradicting Lemma 6 (the integer $a + b - c$ is symmetric in a, b). In both cases no tiling exists. ☐

Remark 8. The role of the two invariants is essential. For the base-angle-β shape, f_α places the equal sides at $\pi - \beta = \frac{2\pi}{3} + \alpha$, an *even* $\frac{\pi}{3}$-coefficient, so f_α gives an integer value and does not obstruct; f_β does. Each invariant is a necessary condition in its own right (Corollary 5), so using the one that fails is legitimate, not a choice of convenient frame: on any genuine tiling both invariants return an integer.

6. Resolution of the problem

Proof of Theorem 1. Let $p \equiv 3 \pmod{4}$ be prime and suppose some triangle is p-tiled. By the classification (Section 2) the tile is in one of the listed types, and every type other than a $2\pi/3$ tile on a non-equilateral triangle excludes a prime $\equiv 3 \pmod{4}$ by an existing theorem [11, 1, 4, 2, 3, 5]. For a $2\pi/3$ tile, Proposition 2 forces ABC isosceles, and Theorem 7 excludes that. Hence no such tiling exists. Finally 19 is prime and $\equiv 3 \pmod{4}$, so no triangle is cut into 19 congruent triangles. ☐

7. Verification

The finite and numerical claims are checked by the scripts in the repository (exact arithmetic). `verify_shapes.py` reproduces the eleven-shape enumeration of Section 2 and the values N_0 of Section 3. `verify_invariant.py` confirms Lemma 4 ($C_f(t) = \pm(c + a - b)$ over all orientations of T), validates Lemma 3 by computing both sides on explicit subdivision tilings, exhibits the cancellation (4) at a non-edge-to-edge incidence, and confirms Lemma 6 has no counterexample among primitive 120°-triples up to $c \approx 9 \times 10^6$. Lemma 6 is also formalized and machine-checked in Lean 4 with Mathlib (`lean/Erdos634.lean`, theorem `k_not_dvd_sum_sub`); the proof term depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`.

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